



Accurate charge densities from powder X-ray diffraction – a new version of the Aarhus vacuum imaging-plate diffractometer

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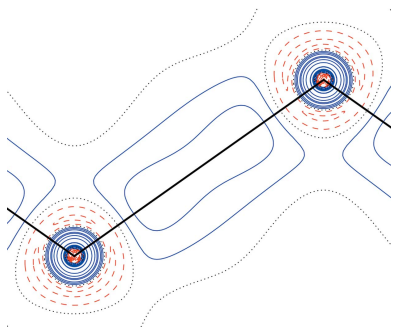
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In recent years powder X-ray diffraction has proven to be a valuable alternative to single-crystal X-ray diffraction for determining electron-density distributions in high-symmetry inorganic materials, including subtle deformation in the core electron density. This was made possible by performing diffraction measurements in vacuum using high-energy X-rays at a synchrotron-radiation facility. Here we present a new version of our custom-built in-vacuum powder diffractometer with the sample-to-detector distance increased by a factor of four. In practice this is found to give a reduction in instrumental peak broadening by approximately a factor of three and a large improvement in signal-to-background ratio compared to the previous instrument. Structure factors of silicon at room temperature are extracted using a combined multipole–Rietveld procedure and compared with *ab initio* calculations and the results from the previous diffractometer. Despite some remaining issues regarding peak asymmetry, the new diffractometer yields structure factors of comparable accuracy to the previous diffractometer at low angles and improved accuracy at high angles. The high quality of the structure factors is further assessed by modelling of core electron deformation with results in good agreement with previous investigations.

1. Introduction

Traditionally, accurate X-ray diffraction data for the determination of the electron-density distributions (EDDs) in crystals have been collected using single-crystal X-ray diffraction (SCXRD; Jørgensen *et al.*, 2014; Coppens *et al.*, 2005; Koritsanszky & Coppens, 2001). However, for SCXRD measurements there are some pertinent issues with specific systematic errors that may affect data quality. These include extinction, absorption and scaling between subsequent frames in the data collection. Extinction is present in highly perfect crystals and effectively decreases the measured intensity of strong, low-order reflections, which are also the most important reflections for determination of the EDD as they contain most information about the valence EDD involved in chemical bonding. Extinction is generally difficult to model in a least-squares refinement, and thus it introduces systematic errors into the determined EDD (Schmøkel *et al.*, 2013). Absorption is particularly important for heavy elements in dense solids. It can be modelled using either face indexing of the crystal or an empirical absorption correction using a highly redundant data set. Uncorrected absorption may also intro-



duce systematic errors in the modelled EDD. The usual remedy for minimizing extinction and absorption effects is to measure data at high energy on very small crystals. Scaling between subsequent frames is used to remove fluctuations in the measured intensity during measurement due to, for example, beam instability, beam inhomogeneity or imperfect crystal centring. However, introduction of a large number of scale factors is inherently an empirical process that may introduce additional systematic errors into the reduced data set because of correlations, in addition to decreasing the significance of the measured Bragg reflections (Jørgensen *et al.*, 2012). The three issues of extinction, absorption and scaling can be at least partially solved by using powder X-ray diffraction (PXRD) instead of SCXRD. In PXRD extinction is removed by using tiny crystallites, absorption is relatively easy to correct for because of the homogeneity of the sample, and scaling between frames can be completely avoided. Two major issues are, however, present in PXRD; these are peak overlap and background treatment, which must be addressed in great detail to extract high-quality structure factors from PXRD data.

PXRD has been used for many years in EDD studies employing the maximum entropy method (MEM), which is a statistical reconstruction of the thermally smeared EDD (Sakata & Sato, 1990). In the case of PXRD data a normal structural Rietveld refinement is first carried out to extract structure factors, which are subsequently used for the MEM calculations (Takata *et al.*, 1995). The MEM–Rietveld method has proven useful in extracting many important bonding features, but the thermally smeared nature of the MEM EDD complicates the interpretation (Cargnoni *et al.*, 2004; Iversen *et al.*, 1995). The possible use of PXRD for modelling the static EDD employing the Hansen–Coppens (HC) multipole model (Hansen & Coppens, 1978) was initiated by Svendsen *et al.* (2010), who obtained the static EDD from the HC model in diamond based on accurate synchrotron PXRD data measured by Nishibori *et al.* (2007). The PXRD multipole refinement introduced a new method of partitioning overlapping reflections based on the HC model, and this reduced bias towards the independent-atom model (IAM), which is typically used in MEM–Rietveld studies.

Following these initial results, we initiated a research program with the goal of obtaining high-quality PXRD data by performing the diffraction measurements in vacuum and thereby removing the air scattering. In this paper, we briefly review the method of diffraction in vacuum and the results obtained from our custom-built Aarhus vacuum imaging-plate diffractometer (AVID). Afterwards, we present a new and modified version of the diffractometer, producing data of outstanding accuracy with significant improvements compared with the previous version of AVID.

2. Diffraction in vacuum

In order for PXRD to be a viable alternative to SCXRD for measuring highly accurate structure factors and modelling EDDs potentially with subatomic resolution (Fischer *et al.*,

2011), the background has to be minimized, *i.e.* the signal-to-noise ratio must be maximized, and the peak overlap must be limited. The background in a diffraction experiment can be roughly divided into two classes: (i) intrinsic effects such as Compton scattering, thermal diffuse scattering and fluorescence from the sample, and (ii) external effects from the sample environment such as air scattering (100 mm of air scatters roughly the same as 0.1 mm solid) and scattering from the sample container (*e.g.* a glass capillary). One possible way of minimizing the background is to use analyser crystals to remove inelastic scattering (Hodeau *et al.*, 1998), but this is a time-consuming technique and requires a dedicated setup on an accurate goniometer. Additionally, it introduces additional scaling between different detector angles as the synchrotron beam may vary over time. Another method is to remove all air scattering by performing the diffraction measurements in vacuum. Recently, the first version of the AVID was reported (Straasø *et al.*, 2013, 2014). By performing PXRD in vacuum, the external background signal can, in the absence of fluorescence, be limited to (i) scattering from the sample container, which can be measured individually and subtracted, (ii) Compton scattering from the sample, and (iii) thermal diffuse scattering, which can be minimized by carrying out data collection at low temperature (Wahlberg, Bindzus, Christensen *et al.*, 2016). The diffractometer had a sample-to-detector distance of 300 mm (Straasø *et al.*, 2013) and used a Fuji BAS SR-2040 imaging-plate (IP) detector positioned on a cylindrical path at the given distance. IPs are nearly ideal as detectors for this setup because of their high detective quantum efficiency, large dynamic range, high spatial resolution and good flexibility (Amemiya, 1995). The drawbacks of using IP detectors are that they have slow readout, natural decay of the phosphor and small errors in position introduced during digitization, as described below.

After digitization, the diffraction data are present as sections of Debye–Scherrer (DS) cones projected onto a cylindrical detector. To produce a suitable one-dimensional PXRD pattern, the two-dimensional data must be azimuthally integrated. This is done using the footprint of a DS cone on a cylindrical detector, as introduced by Straasø *et al.* (2013), transformed into the flat IP coordinate system. Furthermore, a procedure for estimating uncertainties is used, where we assume that each point along the integration arc constitutes an independent measurement of the intensity at a given 2θ value, meaning that we can use simple sample statistics to estimate the standard deviation on the integrated intensity (Wahlberg, Bindzus, Bjerg *et al.*, 2016). All data presented here were collected with various versions of AVID at the P02.1 beamline at PETRA III (Dippel *et al.*, 2015).

3. Results obtained with the previous version of AVID

3.1. Extracted structure factors

The scope of the AVID project is to obtain the best possible PXRD data for experimental EDD modelling. As mentioned above, EDDs are typically determined based on modelling of

structure-factor amplitudes obtained from an SCXRD experiment. Thus, a prerequisite for accurate EDD modelling from PXRD is that the extracted structure factors are of high quality. For highly perfect crystals such as diamond and silicon kinematic scattering cannot generally be assumed for large single crystals, so accurate structure factors have in the past been measured using the dynamical *Pendellösung* technique, which has become the grand standard for accurate structure factors (Kato, 1988; Cummings & Hart, 1988; Takama *et al.*, 1990). Fig. 1 shows the comparison of structure factors from the *Pendellösung* method with the extracted structure factors from the previous AVID PXRD data on diamond at room temperature, and on silicon at room and liquid-nitrogen temperatures (Bindzus *et al.*, 2014; Wahlberg, Bindzus, Bjerg *et al.*, 2016). All structure factors are static, *i.e.* corrected for isotropic thermal vibration (the occupied sites in diamond and silicon have cubic symmetry). The agreement for all three data sets is excellent with agreement factors, $R = \sum |F_{\text{PEND}} - F_{\text{AVID}}| / \sum F_{\text{AVID}}$, of 0.86, 0.25 and 0.57%, respectively. The *Pendellösung* method can measure accurate data, but owing to the very time-consuming nature of the method only structure factors to modest resolution are available. In addition to the good agreement with the *Pendellösung* data, structure factors of much higher order can be collected quickly (a few hours) using PXRD. The high-order structure factors are essential for accurate modelling of thermal vibrations and core electron deformation (Bindzus *et al.*, 2014).

3.2. Electron density modelling and core deformation

As a general notion of chemistry, core electrons are assumed to be inert and therefore not involved in chemical bonding. In line with this, the widespread HC multipole model treats the electron density of the core as fixed and only

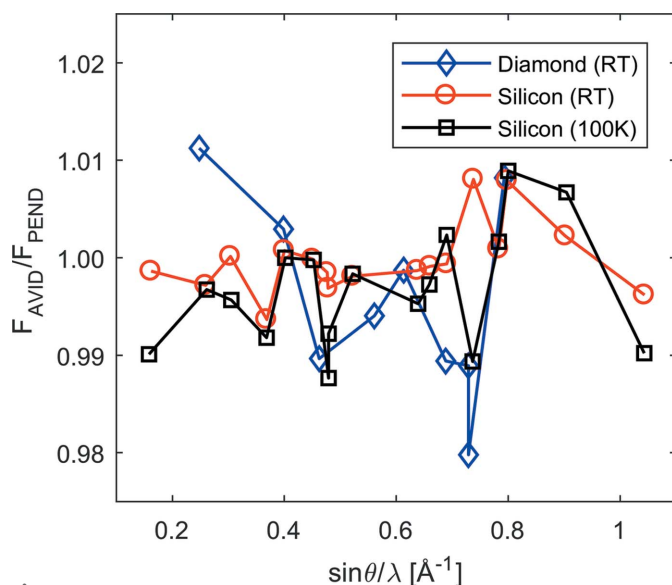


Figure 1
Comparison of structure factors from the *Pendellösung* method with the structure factors extracted using PXRD with AVID. Data are taken from Bindzus *et al.* (2014) and Wahlberg, Bindzus, Bjerg *et al.* (2016). The lines serve as guides to the eye.

polarization of the valence electrons is modelled, so the pseudo-atom electron density is given as (Hansen & Coppens, 1978)

$$\rho_{\text{at}}(\mathbf{r}) = P_c \rho_c(r) + P_v \kappa^3 \rho_v(\kappa r) + \sum_{l=0}^{l_{\text{max}}} \kappa'^3 R_l(\kappa' r) \sum_{m=0}^l P_{lm\pm} d_{lm\pm}(\theta, \varphi), \quad (1)$$

where P_c and P_v are the spherical core and valence populations, respectively, and the aspherical valence is modelled by real spherical harmonics $d_{lm\pm}$ with population parameters $P_{lm\pm}$. Radial expansion and contraction of the valence electron density is modelled by κ and κ' for the spherical and aspherical part, respectively.

Despite the merits of the HC model, theoretical studies have shown that the EDD in the core region is also affected by bond formation, and this can be modelled by extending the HC model to include deformations of the core electron density (Fischer *et al.*, 2011). This model has been coined the extended Hansen–Coppens (EHC) model and the pseudo-atom electron density is given as

$$\rho_{\text{at}}(\mathbf{r}) = \sum_{n=1}^N [P_n \kappa_n^3 \rho_n(\kappa_n r) + \sum_{l=0}^{l_{\text{max}}} \kappa_n'^3 R_{nl}(\kappa_n' r) \sum_{m=0}^l P_{nlm\pm} d_{nlm\pm}(\theta, \varphi)], \quad (2)$$

where the first sum runs over all shells in the atom. In the experimental studies presented here, it has been sufficient to include spherical deformation of the inner shells.

Regarding the physical interpretation of the core deformation, Batke & Eickerling (2013) showed that in the case of d^0 transition-metal complexes, the core deformation could be attributed to the nodal structure of the d electrons involved in bonding, which is not modelled by the nodeless Slater-type radial functions typically employed when using the HC model. Using the EHC model these deformations could be modelled.

To assess the core deformation experimentally, accurate data over a large range of $\sin \theta / \lambda$ are needed, which is indeed the case for the AVID data. Recent AVID PXRD studies have allowed modelling of the core electron deformation in diamond, silicon and cubic boron nitride (Bindzus *et al.*, 2014; Wahlberg *et al.*, 2015; Wahlberg, Bindzus, Bjerg *et al.*, 2016). Here the results for diamond and silicon and the methods employed will be briefly reviewed. To model the electron density, the Rietveld method as implemented in JANA2006 was used (Petricek *et al.*, 2014; Rietveld, 1969). The HC multipole model was used directly in the pattern fitting to obtain the best possible extraction of overlapping diffraction peaks, for which the intensity is partitioned based on the employed model. To successfully deconvolute thermal motion from the thermally smeared electron density, a Wilson plot was used in the case of monoatomic solids. Such plots compare the extracted structure factors with theoretically calculated static structure factors to obtain an isotropic displacement parameter, U_{iso} . The resulting U_{iso} was then used as a fixed value in the subsequent Rietveld cycle, and this was performed iteratively until convergence was obtained. After conver-

gence, a set of structure factors was obtained, and subsequent HC and EHC multipole refinements with fixed scale factor and displacement parameters were performed on these structure factors to reduce the number of parameters refined simultaneously and thus increase the robustness of the final model.

In diamond a contraction of the core shell, $\kappa_c = 1.006$ (1) and $P_c = 1.987$ (4), was observed in excellent agreement with theoretical predictions (Bindzus *et al.*, 2014). The origin of the core contraction is that the $2s$ orbital possesses a radial node, which means that a deformation of the valence electrons upon covalent bond formation will also affect the EDD close to the nuclei. In this respect, it is important to think of the refined core deformation parameters as related to the EDD in the core *region* and not necessarily the EDD in the core *orbitals*, since both valence and core orbitals contribute to the total density in the core region.

In silicon two core shells, K and L , are present, and if possible these should be modelled separately. This was possible for data collected at 100 K, whereas only the L -shell deformation could be modelled at room temperature as Bragg peaks were only available at lower $\sin \theta/\lambda$ owing to the higher temperature. The refined core deformations differ slightly between the two temperatures, where the 100 K model gives an increase in the core density combining contributions from all shells, whereas the room-temperature model gives a decrease. The increase in the 100 K model is similar to what is observed in diamond (Wahlberg, Bindzus, Bjerg *et al.*, 2016). Besides the mentioned studies on diamond and silicon, the EDD of cubic boron nitride (cBN) and copper have also been studied with AVID (Wahlberg *et al.*, 2015; Wahlberg, Bindzus, Christensen *et al.*, 2016). For cBN, it was possible to determine the deformation of the core EDD, even without the use of Wilson plots, which can only be rigorously used for monoatomic solids.

The sample cooler used in all low-temperature experiments and also implemented in the new version of AVID was discussed by Wahlberg, Bindzus, Christensen *et al.* (2016). It works by injecting helium gas cooled by liquid nitrogen onto the sample through a thin nozzle positioned a few millimetres from the sample and beam position. Dynamic pumping and the small flow of helium combined with the low scattering power of helium ensures that the background signal is only slightly increased. The temperature obtained from the cooling device was calibrated against the known lattice thermal expansion and Debye temperature of copper, showing a sample temperature of 95 (3) K. In addition, the anharmonicity of copper was investigated using the Gram–Charlier expansion showing an increased nuclear probability density in the directions away from the nearest neighbours. This illustrates another important point, namely that very accurate PXRD data not only give access to the static EDD, but also the dynamic properties of crystals including anharmonic effects.

It may be noted that following our development of AVID, normal HC multipole modelling has now also been performed using PXRD data collected at the more traditional powder diffractometers at beamlines BL02B2 and BL44B2, SPring-8,

using similar IPs and combining multiple diffraction patterns, but without performing diffraction in vacuum (Kasai & Nishibori, 2016, 2017).

4. A new version of the Aarhus vacuum imaging-plate diffractometer

In order to improve the data quality obtainable from PXRD, thus permitting studies of materials with increasingly complex structure and chemical bonding, a new version of AVID has been designed and commissioned.

4.1. Principle of increasing sample-to-detector distance

The major change to the new AVID design is an increase in the sample-to-detector distance from 300 to 1200 mm. This change is based on the principle that the radial dependences of coherent and incoherent scattering are very different. The incoherent scattering, mainly Compton scattering, only contributing to the background, is scattered as a sphere radiating away from the sample, meaning that the intensity must decrease as R^{-2} . Neglecting divergence, the coherent scattering, including the Bragg scattering, however, radiates away from the sample in DS cones, meaning that the intensity in a point decreases as R^{-1} . Thus, increasing the sample-to-detector distance from R_1 to R_2 should improve the signal-to-background ratio by a factor of R_2/R_1 , in this case a factor of four. This should significantly improve the treatment of the background signal in the PXRD data.

The other important issue present in PXRD, peak overlap, can also be significantly reduced by employing an increased sample-to-detector distance, as minimizing the peak broadening will decrease the effect of partial peak overlap. Recently, both Straasø *et al.* (2013) and Dippel *et al.* (2015) have listed the important contributions to peak broadening in a PXRD experiment. For samples with negligible intrinsic broadening due to size and strain effects and when using a cylindrical detector, the major contributions to peak broadening are (i) the capillary diameter, (ii) the beam height, which is effectively the capillary diameter if the capillary is fully embedded in the beam, (iii) the pixel size including point-spread function, and (iv) the divergence of the incoming beam. Straasø *et al.* (2013) reported simple expressions for these broadening contributions, and it is seen that all of them decrease with sample-to-detector distance as R^{-1} , except the divergence contribution which is independent of distance. If divergence of the beam is negligible, this means that peak broadening can be reduced to one fourth by increasing the sample-to-detector distance by a factor of four.

4.2. Technical details

Increasing the sample-to-detector distance by a factor of four, while still performing diffraction in vacuum, presents a serious challenge due to the significantly increased size of the diffractometer. This has been solved by no longer performing the full experiment in vacuum. As X-ray scattering is a very weak process, only a minor portion of the incoming beam is

actually diffracted. This also means that the air scattering present as background in a typical diffraction experiment comes almost solely from the primary beam. Therefore, our new design keeps the primary beam in vacuum until essentially blocked by a beam attenuator (the beam is attenuated by approximately eight orders of magnitude). The highly attenuated primary beam, used for directly determining the zero position on the detector, and the scattered beams then exit the vacuum chamber through a Kapton window and travel the secondary flight path at ambient conditions, giving rise to negligible air scattering. The increase in detector area also gives rise to a change regarding the IP detector, as no IP scanner exists that is suitable for scanning an IP of 1250 mm required for obtaining diffraction data up to an angle of $2\theta = 60^\circ$ at a distance of 1200 mm, as this diffractometer can cover. Thus, the IP is cut into four pieces with skewed angles at the ends to include some 2θ values in two neighbouring IP strips. This ensures that no 2θ values are missed and makes it possible to stitch the raw data together on the basis of peaks appearing on adjacent IPs. The type of IPs was changed from Fuji SR to Fuji MS, which gives an improvement in sensitivity by roughly a factor of five at the expense of a slightly larger point-spread function (Maddox *et al.*, 2011). This was done since increasing the sample-to-detector distance also gives a decrease in measured intensity per pixel. In the first measurements using this diffractometer with the SR IPs, this led to an intensity of zero or one in many pixels in the background, meaning that linearity of the detector could not be assumed. A schematic drawing of the new version of the AVID is shown in Fig. 2. Although the size of the diffractometer

has been increased significantly, it is still portable and can thus be installed on a suitable base at any beamline with suitable beam specifications at a synchrotron-radiation facility.

5. Data quality of the new diffractometer

5.1. Signal-to-background ratio and instrumental broadening

The aim of the diffractometer upgrade was to gain roughly a factor of four in signal-to-background ratio and reduced peak broadening. In Fig. 3, data collected on a silicon standard (NIST 640d) at room temperature at P02.1, PETRA III, Germany [$\lambda = 0.20708$ (6) Å], are shown for the new diffractometer (1200 mm) and its predecessor (300 mm). The data are scaled such that the integrated intensity of the (202) reflection is the same for both data sets.

As the total integrated coherent and incoherent scattering should be the same for both setups, we would expect to see the same background level after this scaling and observe the improved signal-to-background ratio in terms of a sharpening of the peaks due to reduced peak broadening. As evident from Fig. 3, however, we also observe a significant further improvement in background reduction. Reasons for this could be differences in air pressure in the vacuum chamber and different filling fractions of the used capillaries.

The second goal was to reduce the instrumental broadening. A comparison of the full-width at half-maximum (FWHM) as a function of diffraction angle obtained from Rietveld refinement with a pseudo-Voigt profile (Thompson *et al.*, 1987) in JANA2006 (Petricek *et al.*, 2014) is shown in Fig. 4. The

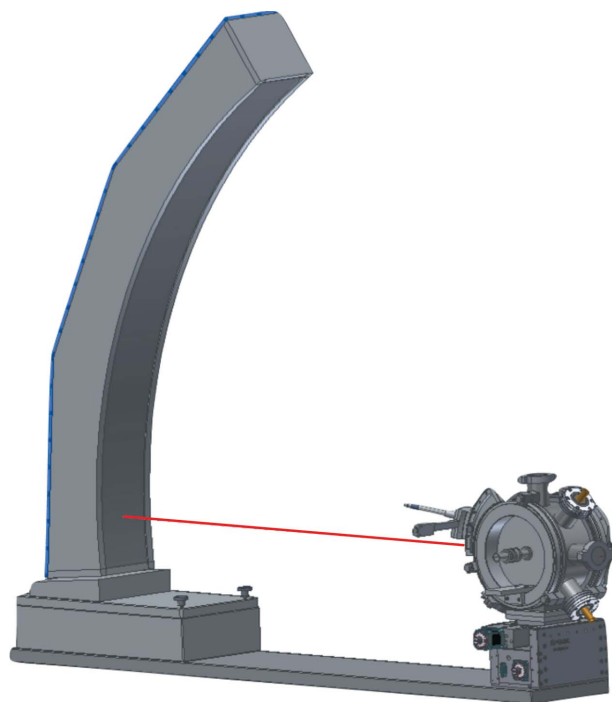


Figure 2

Schematic drawing of the new vacuum diffractometer with a sample-to-detector distance of 1200 mm. The direct, attenuated, beam is shown as a thin red line. Two detectors for various auxiliary beam analyses are also shown in the schematic.

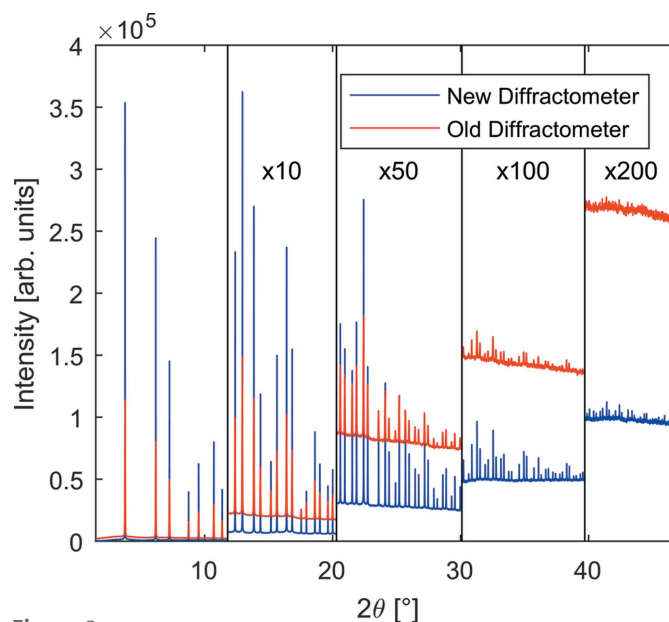


Figure 3

Comparison of data collected on NIST silicon powder at room temperature from the new version (1200 mm) with the old version (300 mm) of the diffractometer. Data are shown corresponding to the angular range covered by the first three IPs in the new setup. Both data sets are raw integrated data without any background subtraction performed. The data have been scaled to have the same integrated intensity of the (202) reflection.

refined parameters are shown in Table S1 of the supporting information. The improvement in instrumental resolution is slightly larger than a factor of three at low angles and slightly less than a factor of three at higher angles, in good agreement with the predicted improvement taking the new IPs into account. The similar angular dependence of the instrumental broadening in the two setups suggests that the geometric broadening is limited to a degree where we observe the effects of the divergence of the synchrotron beam. The improved signal-to-background ratio and instrumental broadening clearly leads to well resolved Bragg peaks at higher angles compared to the old diffractometer.

5.2. Scanner position correction and asymmetry

Until now, we have assumed symmetric peak profiles such as a pseudo-Voigt profile. A thorough inspection of data, however, shows that some asymmetry is present. To model this asymmetry, we have performed single peak fitting with a split pseudo-Voigt profile, which uses two half-width at half-maximum (HWHM) parameters and two Lorentzian mixing parameters, one for each side of the peak. The difference in these parameters can be used as a measure of the degree of asymmetry in the peak profiles, and is shown in Fig. 5.

It is clearly seen that the profiles consistently become broader and more Lorentzian at higher angles on both sides of the peak, but it is also evident that asymmetry is present at all angles considered, and there is no tendency for the asymmetry to decrease with angle, as would be the case for axial divergence asymmetry (Finger *et al.*, 1994). Instead, the increase in peak width and the remaining peak asymmetry with increasing angle could suggest that we in fact observe broadening due to

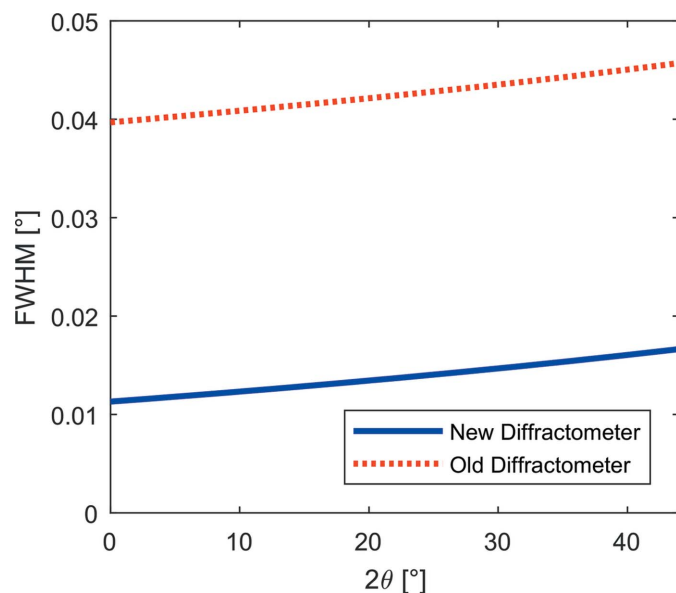


Figure 4 Instrumental broadening from the new and the old diffractometer. The FWHMs from a Rietveld refinement on data from NIST silicon powder in JANA2006 are used as measures for the instrumental broadening and are shown up to 44°, which is the data range included in the Rietveld refinement.

the wavelength profile of the incoming radiation, which might not be completely symmetric. This asymmetry only becomes observable in the case where geometric sample-broadening effects are absolutely minimal. It should be noted that asymmetry is present even when only looking at data from a one-pixel-wide line through the DS cone apex. This means that the observed asymmetry is not an artefact of the integration path, which resembles the real footprint of the DS cone as closely as possible to avoid introducing asymmetry in the azimuthal integration procedure.

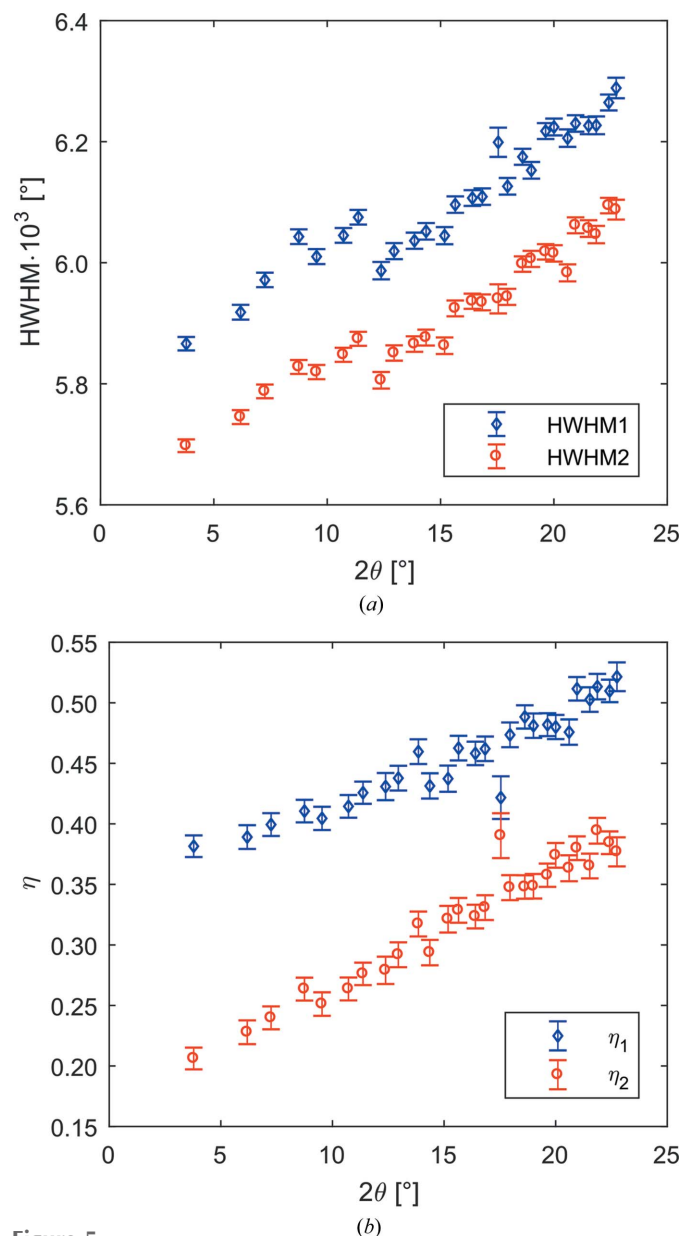


Figure 5 (a) Half-width at half-maximum (HWHM). (b) Lorentzian mixing parameter (η) for left and right sides of reflections as determined from single peak fitting using a split pseudo-Voigt profile on data from NIST silicon powder. Index 1 refers to the low- 2θ side of the peaks. The small drop in HWHM at around 12° coincides with the change from IP1 to IP2 and may be due to some unresolved misalignment of the IPs in the scanner. Only part of the data range is shown as single-peak fitting becomes less accurate at higher angles.

Another issue present when peak broadening is minimized is the effect of small errors during the digitization process. The IPs are digitized by a scanner (GE Typhoon FLA 7000) that excites the metastable states in the IP *via* a laser and a photomultiplier tube that reads out the intensity of emitted photons. This is done in increments of 25 microns, which is the pixel size of the data after scanning. However, during this process small errors in the observed positions are introduced. This is observed in terms of a mis-positioning of the measured diffraction peaks. The error amounts to a few pixels, and since one pixel in this setup corresponds roughly to 1.2 millidegrees, the error in position is a considerable fraction of the FWHM, making a full pattern refinement, which assumes no error in scattering angle, impossible if no correction is performed.

The correction is performed using a LaB₆ standard (NIST 660b). LaB₆ is used owing to the larger primitive unit cell compared to silicon, thus giving rise to more diffraction peaks. A narrow strip of each IP is integrated, and single-peak fitting is performed to find the observed diffraction angles. IP1 and IP3 are positioned on the same side of the scanner in two consecutive scans and should therefore give a similar error curve, if the error is reproducible. The diffraction peaks on IP4 and the final part of IP3 are too weak to perform reliable single-peak fitting. In Fig. 6, the observed deviation from the calculated position is shown for the first three IPs. The curves are very similar, but there is clearly a difference between IP2 and IP3 in the low pixel values. IP1 and IP3, which are scanned from the same position, show very similar trends in the area, where peaks are found on both IPs. From these data, two correction curves are made, one for each half of the IP scanner, by using a smoothing spline to get a smooth curve with estimated correction values at all positions in the scanner. This correction is then applied to the raw data before any integration is performed.

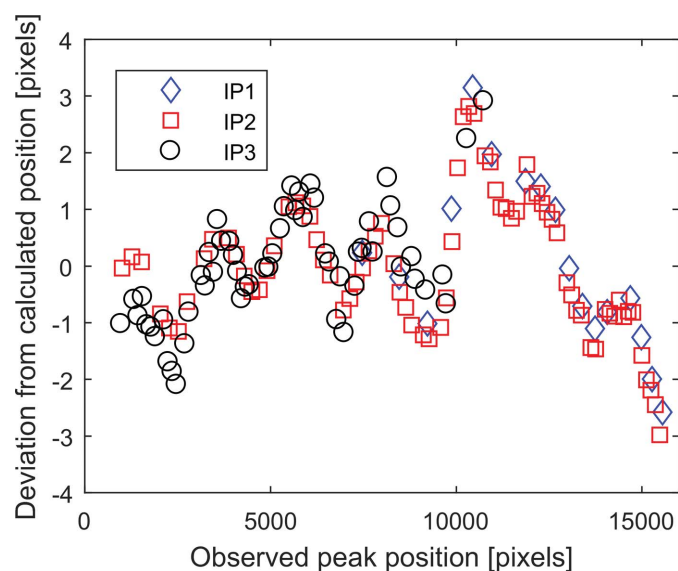


Figure 6

Deviation from the calculated peak position for a LaB₆ standard sample as a function of measured peak position. There is clearly a difference between IP2 and IP3 in the low pixel range, which makes different corrections for different IPs necessary.

With the extremely narrow peaks in the present data set, a further correction must be performed to remove the deviations of around 0.5 millidegrees still present after the LaB₆ correction due to the finite number of diffraction peaks and errors in the peak-fitting procedure. This requires knowledge of the unit cell for each sample, which is, however, easily determined by in-house powder diffraction or from single-peak fitting on the uncorrected data, assuming that the error is distributed around zero. An important note here is that the

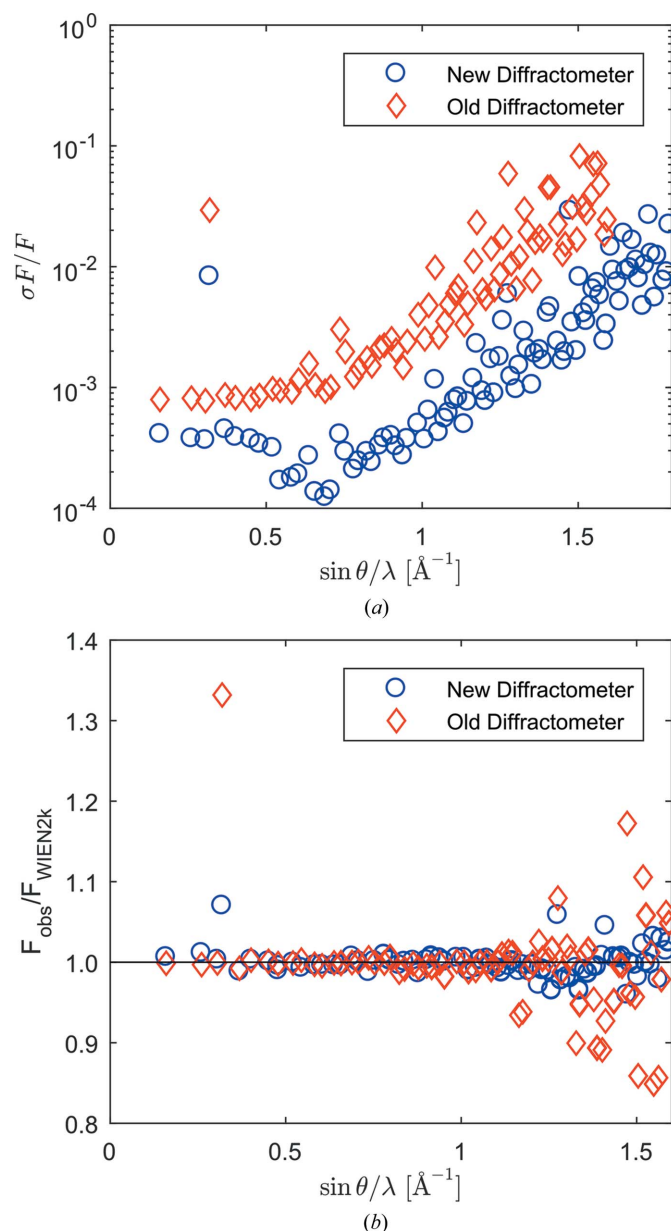


Figure 7

(a) Significance of extracted structure factors of silicon from the new and old AVID. (b) Comparison of ADP-corrected extracted structure factors of silicon from the new and old AVID with calculated static structure factors from WIEN2k. Only reflections for which the structure factors could be extracted for both data sets are shown. The largely deviating reflection around $0.3 \text{ \AA}^{-1}$ is the weak 'forbidden' (222) reflection. The reflection is only forbidden if the atomic densities are treated spherically. The structure factors from the old diffractometer are given in the supporting information of Wahlberg, Bindzus, Bjerg *et al.* (2016).

Philip Coppens tribute

Table 1

Refined multipole parameters and topological properties at the bond critical point from the EHC model on silicon with structure factors extracted from the old and the new AVID and from theory.

Parameters from the old AVID and from theory are taken from Wahlberg, Bindzus, Bjerg *et al.* (2016).

	Old 100 K	Old 300 K	Theory	New 300 K
Cut-off ($\text{\AA}^{-1}$)	1.7	1.3	1.8	1.7
Scale	0.9936	1.0026	1.0	0.9977
U_{iso} ($10^{-4} \text{\AA}^2$)	31.83	56.20	0.0	61.03
P_{M}	4.05 (1)	4.00	4.055	4.04 (2)
κ_{M}	1.029 (8)	0.98 (2)	0.979	1.08 (1)
κ'_{M}	1.42 (4)	1.24 (4)	1.002	1.47 (5)
P_{32-}	0.33 (2)	0.39 (4)	0.424	0.24 (2)
P_{40}	-0.12 (2)	-0.21 (5)	-0.201	-0.19 (5)
P_{L}	7.95 (1)	8.000 (2)	7.978	7.96 (2)
κ_{L}	1.013 (1)	0.998 (6)	1.0085	1.007 (1)
P_{K}	2.00	2.00	1.966	2.00
κ_{K}	0.9482 (8)	1.00	1.023	0.952 (7)
ρ_{BCP} ($\text{e \AA}^{-3}$)	0.61	0.61	0.56	0.60
$\nabla^2 \rho_{\text{BCP}}$ ($\text{e \AA}^{-5}$)	-3.5	-4.1	-3.1	-3.1

final correction must be performed with the same peak profile as the one intended for use in the full pattern refinement, since if asymmetry is present the definition of peak position may vary between different implementations of asymmetric peak profiles.

5.3. Structure-factor quality and electron-density modelling of silicon

Despite the current limitation in peak profile, structure factors have been extracted from the new AVID using the same method as Wahlberg, Bindzus, Bjerg *et al.* (2016) using a Wilson plot for silicon at room temperature. In both cases, the uncertainty of a point in the powder pattern is estimated from sample statistics on the two-dimensional detector, and the uncertainties on the extracted structure factors are determined from error propagation in JANA2006 as described in the supporting information of Wahlberg, Bindzus, Bjerg *et al.* (2016).

The significance of the structure factors is greatly improved, as seen in Fig. 7(a), with an improvement of almost an order of magnitude for many reflections due to the lower noise level obtained with the new AVID. It is also seen that structure factors can be extracted to higher $\sin \theta/\lambda$ values. In Fig. 7(b), the extracted and ADP-corrected structure factors (where ADP is atomic displacement parameter) from the new and the old diffractometer are compared with static structure factors calculated using WIEN2k (Blaha *et al.*, 2008; Wahlberg, Bindzus, Bjerg *et al.*, 2016). At low angles, the structure factors from the new diffractometer are deviating slightly more from theoretical values, probably because of the limitation in peak profile, whereas the agreement is improved at higher angles compared with the old diffractometer. This gives promise that the new AVID will provide data of high quality over a large range of $\sin \theta/\lambda$ suitable for modelling of core EDD features and fine details of thermal vibrations such as anharmonicity.

A preliminary EDD modelling was performed on the extracted structure factors with scale factor and isotropic displacement parameter fixed at the values obtained from a Wilson plot. In Fig. 8 the residual density is shown in the (110) plane after modelling using the HC and the EHC model, respectively. A large residual is present in the core region in the HC model, which completely disappears in the EHC model. Only very low residuals comparable to the noise level are present in the valence region in either model. Refined parameters in the EHC model and topological properties at the bond critical point are shown in Table 1 and compared with the results obtained from the old AVID and from theory. The agreement between the present model and the model from the old 100 K data, which were available to the same resolution, is very good. With the exception of the expansion/contraction parameter of the inner shell, the agreement with

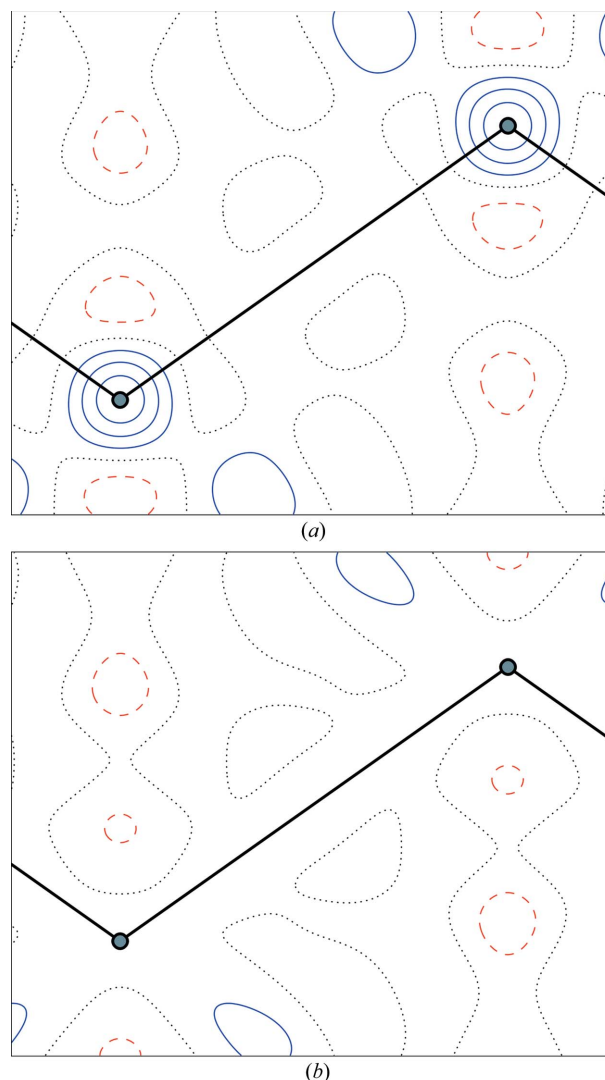


Figure 8
Residual density maps of silicon in the (110) plane based on all reflections below $\sin \theta/\lambda = 1.0 \text{\AA}^{-1}$. (a) HC model. (b) EHC model. Both models are based on extracted structure factors with fixed scale factor and isotropic displacement parameter based on a Wilson plot. Positive (solid blue), negative (dashed red), zero (dotted black) contours are shown. Contour levels are drawn at $0.05 \text{ e \AA}^{-3}$.

parameters from refinement against theoretical structure factors is also very good. The static model deformation density and Laplacian from the EHC model are shown in Fig. 9 and show the charge accumulation in the covalent bonding region in addition to the changes in electron density in the core region from the EHC model.

A more thorough analysis of the EDD at both room temperature and low temperature and a comparison of extracted structure factors with *Pendellösung* structure factors will be performed when an improved peak profile has been implemented in a Rietveld refinement package suitable for performing combined Rietveld and HC multipole modelling.

6. Conclusion

We report a new version of our custom-built Aarhus vacuum imaging-plate diffractometer, AVID, which is capable of

measuring very high quality PXRD data. By increasing the sample-to-detector distance from 300 to 1200 mm, we have in practice successfully reduced the the instrumental peak broadening by roughly a factor of three, thereby reducing partial peak overlap, and very significantly improved the signal-to-background ratio. Thus, we have successfully minimized the two most important limitations in PXRD compared to SCXRD. Several issues, such as position errors in the scanning process and peak asymmetry, become evident when the peak broadening and background level are reduced to the present level. The scanner position error can be corrected for, and the asymmetric peak profiles can be implemented in a Rietveld refinement. Despite this current limitation in peak-profile modelling, structure factors of comparable quality to those obtained from the old version of AVID can already be extracted. In particular, the structure factors at high resolution are of improved accuracy. The improvements discussed here open the way for EDD modelling of materials with increasingly complex structures and chemical bonding. For extended inorganic structures in particular, PXRD should be considered a valuable alternative to SCXRD for EDD studies as these are often severely affected by extinction and absorption, the effects of which are less pronounced in PXRD. Furthermore, these extended inorganic structures often exhibit small unit cells of relatively high symmetry, meaning that peak overlap is less of a serious issue.

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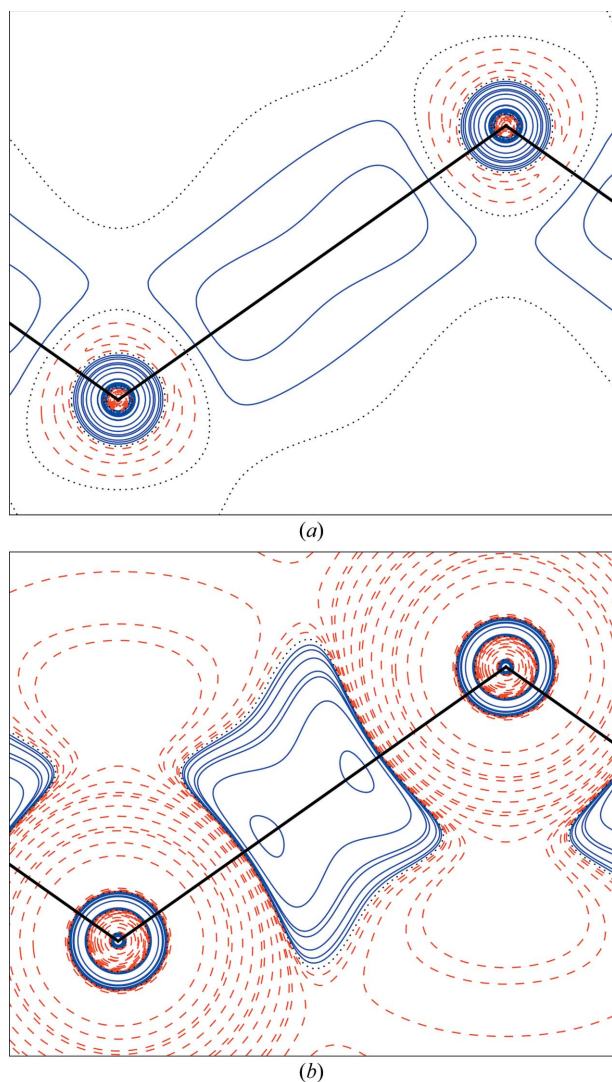


Figure 9 (a) Static model deformation density ($\rho - \rho_{IAM}$) of silicon. (b) Negative Laplacian of the total density ($-\nabla^2\rho$) of silicon. Both maps are drawn with exponential contour levels with bases $\pm 1, 2, 4, 8$ and exponents $-1, 0, 1, 2, 3, 4, 5, 6$. Positive (solid blue), negative (dashed red), zero (dotted black) contours are shown.

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